AUSTRALIAN ACCESS FEDERATION

Access Options for Non-AAF Researchers

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Executive Summary

Research communities and collaborations can reduce their workload by leveraging the policy and trust frameworks, support and technical infrastructure of research and education (R&E) federations to manage access to community applications and data. Some researchers do not have access to this infrastructure, so alternate authentication methods are necessary for a subset of users. This document explores the alternatives where access through an R&E federation is not an option.

Characteristics of the data, computing demands, data sensitivity, size and diversity of the user group ensures each research community has its own unique set of requirements. These characteristics must be factored into the decision to support extra authentication methods. Not all the options will be suitable for every research community.

Federations typically present well-identified users, establish an active relationship with a research organisation, provide a useful range of attributes to assist the application to make authorisation decisions and may signal multi-factor authentication when it occurs. The alternatives dilute one or more of these dimensions and each research community must assess whether these compromises are acceptable. Sometimes shortcomings can be addressed through extra verification and workflow steps.

Federation identity quality is maintained through trust frameworks that address not only technical interfaces but also policy, governance and legal requirements. The alternatives discussed offer a degree of technical interoperability but rarely address the other elements. This leaves the research community and the collaboration operator to resolve the policy and governance challenges and sometimes this means they must operate components of the technical solutions and supporting services.

Federated Identity and Trust in Global Research Collaboration

Let's suppose a medical researcher from the University of Sydney (USyd) wants access to some data stored in the National Institutes of Health (NIH) Data Portal in the US. The researcher can log in to the Data Portal with their USyd account without any prior agreement and even without prior knowledge of the NIH. How is this possible?

Technology plays a role. Common technical standards, shared metadata, cryptography and assurance signalling all enable the NIH to know the researcher is a bona fide individual from an organisation involved in research and scholarship. But technology alone is not sufficient.

The technology works because of a common policy framework, even in the absence of direct agreements, each party can trust that the other follows similar practices. Stacked trust frameworks mean USyd and NIH follow similar rules of their respective national federations. The AAF and InCommon federations also follow similar practices outlined in the eduGAIN Policy Framework Declaration.

This layered policy approach has made it possible to scale interoperability in the global research and education community. Other approaches would be unworkable with more than 80 national federations, 6000 identity providers and 3700 services.

Evaluation Criteria for Candidate Solutions

The evaluation criteria must be defined before assessing the appropriateness of candidate solutions. This section lays out the key factors used to evaluate the alternatives identified later in the document.

1. Does the option minimise the involvement of the research community in managing identities? (i.e. the solution leverages processes and technologies of the provider to reduce the identity and access management workload for the research community)
2. Does the credential issuer verify the identity of the subject against government identity documents when issuing new credentials?
3. Does the identity provider assert a relationship between the user and a recognised research organisation?
4. Does the identity provider assert attributes to identify the user and assist in making authorisation decisions (e.g. first name, surname, email address, home organisation, affiliation, identity assurance level, authentication assurance level (MFA))

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5. Does the identity provider make timely updates to user's attribute data as details change?
6. Does the identity provider operate account lifecycle management services for users (e.g. password and MFA reset, account recovery and helpdesk)?
7. Can the option enable access to multiple research communities (i.e. the user does not need to re-register a new identity to use services in a different research collaboration)?
8. Can shortcomings be overcome by adding further validation processes for a reasonable cost/effort?

Assurance level requirements will vary between research communities since discipline areas and data sets have different risk profiles. Some research communities deal primarily in low-risk data where low identity assurance or even anonymous access is sufficient. Other research communities working with high-risk data sets, containing identifiable human health data, will need stronger identity assurance processes.

The AAF's Trust and Identity Policy Development Kit[1] helps research communities assess the inherent data and service-related risks and develop a policy suite to mitigate and manage these risks. The policy requirements of a research community play a significant role in determining whether the alternate solutions are fit-for-purpose for that specific community. Creating these policy statements first will make it easier to determine which of the alternatives can meet the needs of the research community.

Federated Identity – the Preferred Option

Encouraging organisations to join the relevant research and education federation (AAF or another eduGAIN member) is usually the most preferable option from the point of view of the research collaboration operator. Federation identity providers typically present users with moderate to high levels of identity assurance, establish a direct relationship to a research organisation, and supply a range of attributes to support identification, personalisation, access control and security.

Using federated identities through AAF or eduGAIN contributes the following benefits to the research collaboration:

- leverages the federation rules to ensure good security, privacy, and identity lifecycle management practices
- identity assurance, credential issuing and user lifecycle management processes are the responsibility of the home organisation, not the research collaboration operator
- conveys a tangible relationship between the end user and the research institution – a primary trust marker of the legitimacy of the end user as a researcher
- identity providers are effectively guaranteed to release the attributes needed by the research collaboration (with an escalation path via the federation operator if this happens incorrectly)

This approach is seamless for the researcher as it provides access to all research collaborations and services available through AAF and eduGAIN, not just services connected to a specific research collaboration.

In some circumstances joining the federation may not be an available option or may not be financially justifiable for a research organisation. The remainder of this document will explore potential solutions when joining a federation is not a viable option.

Access Options for Non-AAF Researchers

Many research communities want to include researchers or other stakeholders that do not work for a research organisation that is part of the AAF or another eduGAIN-connected federation. Common examples include stakeholders from:

- local, state and commonwealth government agencies
- commercial research organisations
- non-profit and cultural institutions (e.g. galleries, libraries, and museums)

[1] The AAF T&I Policy Development Kit is based on the AARC Policy Development Kit.

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- community organisations
- independent researchers

The scope of stakeholders could include an entire organisation, a team, or individual researchers. The choice of an option depends on whether an organisation-wide solution is necessary or the scope is just an individual or a small group of users within an organisation. The AAF can provide tailored guidance once the GUARDIANS partners identify those other organisations or individuals with whom they are most likely to collaborate.

Appendix A lists several Australian and international research collaboration examples and identifies the authentication mechanisms they have implemented.

Figure 1 shows a decision tree outlining the current alternatives. These alternatives are discussed in further detail in the following sections.

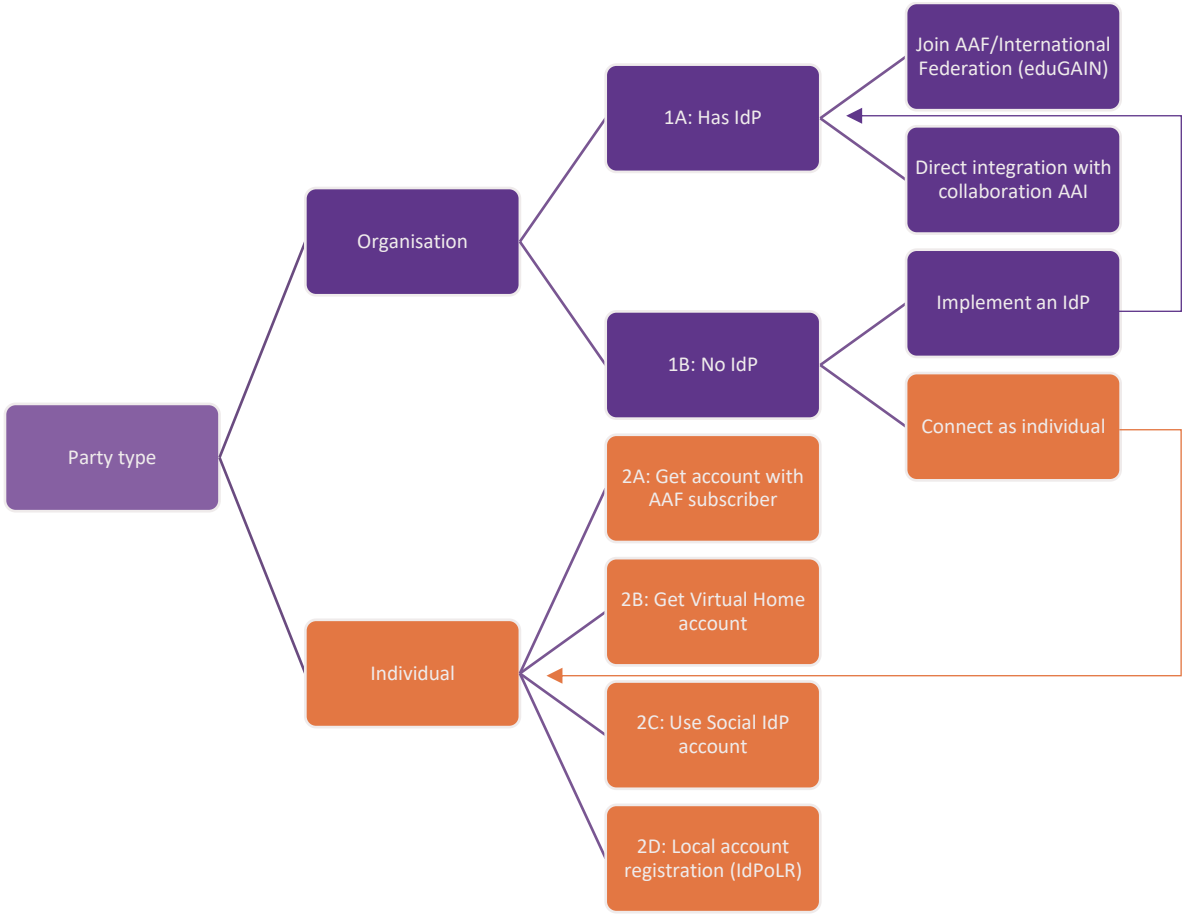

Figure 1: Decision tree for non-AAF user options

Category 1: Organisations

The path for connecting an entire organisation to a research collaboration is heavily influenced by whether they are already invested in a company-wide identity management infrastructure with adequate user management policies and procedures.

Category 1A: Organisations with an Existing Identity Provider

Larger organisations will likely have implemented a platform such as Microsoft Entra ID, Google Workspace or other alternatives for managing user credentials. These offerings typically include a federation ready IdP service.

For organisations with an IdP, there are two primary pathways to participate in research collaborations:

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1. **AAF integration**: Joining AAF is usually the most convenient way forward and brings the advantages outlined in section **Federated Identity – the Preferred Option**.
2. **Direct integration with the research collaboration**: In many cases a point-to-point integration with the organisation's IdP and the research collaboration's AAI will be possible.

Direct integration solves the immediate problem of authenticating and managing access and enables a wider range of organisations to participate but comes with several disadvantages.

- **Scalability challenges.** Direct integration enables the organisation to connect to services integrated with the research collaboration's AAI. The integration work must be repeated for each research collaboration that the organisation wants to join. The organisation will not be able to access more general services (such as AARNet's FileSender) that are part of the AAF but not directly integrated with the research collaboration.
- **Increased governance responsibilities for the research collaboration.** The collaboration assumes the role of a mini-federation operator, responsible for establishing and maintaining its trust framework. This includes agreements, governance and compliance processes for each organisation connected in this way.
- **Increased integration work for the research collaboration.** Each new organisation must be individually onboarded and requires management as identity and access requirements change. Onboarding can be time-consuming, particularly for smaller organisations with lower technical expertise. Attributes will need to be remapped for each technology platform that is directly integrated with the AAI.

Category 1B: Organisations without an Existing Identity Provider

The smaller organisations without existing identity management infrastructure have fewer options for participation in federated research collaborations. Implementing an identity provider is typically the best way forward, but this is a distinct project on its own. The choice of an identity provider service depends on the organisation's specific needs. AAF has a potential solution for small organisations. Commercial solutions (with or without joining AAF) may also be viable alternatives.

Where implementing an organisation-wide IdP is not viable, individual researchers may still be able to participate in the research collaboration. See the Category 2 options for potential solutions for individuals.

Category 2: Individuals and Small Teams

Individuals and small teams within a larger organisation have a more fragmented range of options. Not all of these options will necessarily be available to all researchers. The specific circumstances of the research or research team and the infrastructure and policies of the research collaboration will influence which of the following options are available.

The research collaboration's policies will influence whether it offers one or more of the access methods outlined below. Research communities may not support any of these options if they do not sufficiently address the selection criteria outlined in section **Evaluation Criteria for Candidate Solutions**.

Factors such as how well the user identity is verified when credentials are issued, whether the IdP releases attributes the AAI needs and whether shortcomings can be overcome with extra process at a reasonable cost.

Category 2A: Leverage an existing AAF subscriber

Under some circumstance a researcher may be able leverage a relationship with an existing AAF subscriber to participate in the research community. Universities will typically issue credentials to people with adjunct appointments or with a visitor or contractor relationship to the university. These credentials will typically enable access to all services in the AAF and to services connected to the research collaboration.

The downside of this approach is that the credentials are typically only available while the individual maintains an active relationship with the university. When that relationship ends the credentials are revoked which consequently revokes access to the research collaboration and its services.

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Category 2B: Sponsorship into the AAF Virtual Home

Existing AAF subscribers can create accounts for users they collaborate with in the AAF Virtual Home (VH). The AAF VH is an identity provider used to enable access for individuals collaborating with AAF subscribers on specific projects. Much like Category 2A, it offers access to all services connected to AAF including the research collaboration. The AAF subscriber assumes responsibility for user identity validation, credential issuance and revocation at the conclusion of the research activity or project. VH is intended for individuals for a specific project rather than as an ongoing means of access.[2] VH user onboarding and management can only be delegated to members of an AAF subscribing organisation. VH is not a scalable option as a general approach for enabling access to GUARDIANS services.

Category 2C: Social IdP

Social IdPs such as Google, Microsoft, GitHub and ORCiD offer a quick and convenient way to connect people outside of AAF to a research collaboration. Downsides include low identity assurance and limited attribute release making them less useful for research collaborations dealing with sensitive data sets. The research collaboration may be able to add extra process steps to increase assurance about the user identity but will be restricted to a limited range of attributes to work with than they will receive through AAF or eduGAIN IdPs.

Category 2D: Research Collaboration Local Account (IdPoLR)

Most research collaborations eventually encounter a subset of researchers who are unable to participate through any of the mechanisms discussed above. As the collaboration grows in scope it inevitably implements a means to manage accounts for this subset of users with an Identity Provider of Last Resort (IdPoLR). This approach allows the research collaboration to issue identities for any users that fall through the cracks and provides the flexibility to create as sophisticated a process as necessary to verify user identity before issuing credentials.

An IdPoLR is the identity provider service that supports the registration and authentication of that subset of researchers who are unable to participate through other authentication mechanisms. The IdPoLR will enable access for this group of researchers to only those services connected to the collaboration. The operation of an IdPoLR by a collaboration comes with the additional cost of governance and operational overheads, increased security risk exposure and operational workload.

An IdPoLR can complement or replace options 2A, 2B and 2C and is widely used by research collaborations (e.g. Elixir and the National Institutes of Health (NIH)). With appropriate identity validation processes and policies, this approach ensures that users joining the GUARDIANS research ecosystem are adequately identified before their first access to GUARDIANS' services.

Scoping Effort for Different Options

There are several factors (technical implementation, IT skill, policy and compliance) that determines the effort necessary for an organisation to deploy and connect an identity provider service for their users. A similar degree of effort will be necessary if a collaboration chooses to deploy an IdPoLR.

Social login providers (Google and Microsoft) and the AAF VH service offer the fastest way to integrate. Table 1 outlines anticipated deployment efforts for the options against their inherent identity validation and MFA capabilities.

[2] For further information on VH see: https://support.aaf.edu.au/support/solutions/articles/19000031130-what-is-the-aaf-virtual-home-

Options	Effort estimation (technical/non-technical)	Highest achievable identity assurance[3]	MFA signalling capability
1A.1: Organisations with an Existing Identity Provider – AAF integration	Low/Low	High	Yes
1A.2: Organisations with an Existing Identity Provider – Direct integration	Low/Medium	High	Dependent on IdP selected
1B: Organisations without an Existing Identity Provider	Medium/High	Medium	Dependent on IdP selected
2A: Leverage an existing AAF subscriber	Already enabled with each AAF subscriber	Medium-High	Yes
2B: Sponsorship into the AAF Virtual Home	Already enabled with each AAF subscriber	Medium-High	Yes
2C: Social IdP	Already enabled in the AAI	Low	No
2D: Research Collaboration Local Account (IdPoLR)	Medium/High	Depends on the collaboration's policies and processes.	Dependent on IdP selected

Table 2: Options deployment effort verses identity validation and MFA capabilities.

While most GUARDIANS partners are AAF subscribers, the **Peter MacCallum Cancer Centre** and the **Children's Cancer Institute** have recently joined the AAF (or are in the process of becoming subscribers) and fit the **1A.1: Organisations with an Existing Identity Provider (AAF integration)** option. The AAF has a well-defined and optimised process for on boarding new identity providers. New subscriber's internal processes largely dictate how quickly the organisation can connect to the federation. An organisation's internal processes may include security and risk assessments tailored to their policies, governance and technical environment. Test and production environments can comfortably be deployed, configured and tested inside two weeks but risk assessment and procurement procedures often extend the timeline.

The effort for **1A.1 AAF integration** and **1A.2 Direct integration with the research collaboration** are similar. The later requires repeating the work when joining each new research collaboration.

The technical effort for option **1B: Organisations without an Existing Identity Provider** (to implement an identity provider) and **2D: Research Collaboration Local Account (IdPoLR)** are similar. Option **2D** will require additional effort in the selection of a technology platform and to establish policies for:

- the operation of an IdPoLR that meets SIRTFI criteria[4]
- the onboarding and management of users
- the collection and sharing of personal information with services
- security incident response framework to enable the coordination of incident responses with the collaboration and the services
- other policies necessary for participation in a research collaboration.

[3] Identity validation and verification may also (re)occur as a distinct activity during the data access request phase and as a method of validating identity assertions by institutions.
[4] REFEDS SIRTFI https://refeds.org/sirtfi.

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The policies to operate a research collaboration, such as GUARDIANS, overlap with those necessary for the operation of an IdPoLR. The AAF can provide tailored guidance and support to establish these policy frameworks.

GUARDIANS Project Focused Recommendations

While the GUARDIANS project is in the development and testing phases and has a limited audience, the existing solutions are sufficient for the testing and validating access to the AAI and connected services. Current solutions include the existing AAF subscriber identity providers, VH and the social login providers.

Subsequent project phases should incorporate GUARDIANS' policy and data classification work for specifying identity assurance and authentication assurance (MFA) requirements relative to the risks and sensitivity of GUARDIANS datasets. GUARDIANS may need to extend the methods as the project delivers production services and the diversity of user groups expands.

Conclusion

Steering organisations and researchers to join their local federation is generally the smoothest path where this option is viable. Leveraging the compliance framework of the federation operator and the support resources of home organisations reduces the research collaboration identity management workload and risk profile. There are a range of scenarios where joining a federation is not viable. The risk profile and size of the research community play a major role in determining which of these options are workable in the specific context of each research collaboration. The alternatives typically offer lower assurance about the user identity and a smaller attribute set for services to make authorisation decisions. These limitations can be worked around with extra process steps in some cases.

A collaboration may mix and match the options discussed to onboard different researcher populations. This is influenced by a collaboration's policies, processes and the sensitivity of the data available from a collaboration. These considerations should be formalised and agreed between the collaboration operator and the resource and data partners early in their assessment to eliminate unviable options. This does not preclude the selection of multiple options that can be linked (or limited) to a collaboration's different tiers of data sensitivity, with process and procedures to support step-up identity and authentication assurance for higher risk use cases. The effort to achieve this should be commensurate with the size of the researcher population that will benefit from an option and their community participation.

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Appendix A – Research Community Examples

Appendix A lists example research communities and the authentication options they offer.

Research Community	Region	Research Area	Authentication Methods
ACCESS NRI	Australia	Climate science	• AAF • GitHub • local account (IdPoLR)
Atlas of Living Australia (ALA)	Australia	Biodiversity	• AAF • Facebook • Twitter • Google • local account (IdPoLR)
CADRE	Australia	Humanities and social science	• AAF • eduGAIN • Google • CADRE Guest Login (IdPoLR)
CSIRO Data Portal	Australia	Multiple research domains	• AAF • CSIRO staff account • OPAL (local account IdPoLR)[5] • DataCentral[6]
LDaCA	Australia	Humanities	• AAF • eduGAIN • Google
Life Science Login	Europe	Life science	• eduGAIN/AAF • Apple • GitHub • Google • LinkedIn • ORCiD • LifeScience Hostel (IdPoLR)
TERN	Australia	Environment and climate	• AAF • eduGAIN • Google • GitHub • Microsoft • ORCiD

Note: It is unclear what additional process steps (if any) these communities have implemented to strengthen identity assurance from social identity providers and their IdPoLR.

[5] Australian Telescope National Facility account - https://opal.atnf.csiro.au/
[6] All-Sky Virtual Observatory account

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Appendix B – IdP Attribute Support Comparison

Appendix B shows the attributes that are likely to be received from identity providers commonly used for access to research collaborations

eduGAIN IdPs predominantly assert the same attribute bundle as AAF IdPs, however there may be variations and anomalies between different national research and education federations.

Possible indicates that the identity provider may release an attribute value if the end user has configured the release of that attribute. Relying parties cannot rely on these values to be consistently released.

Attribute	**AAF IdPs**	**AAF VH**	**Google**[7]	**Microsoft**	**GitHub**[8]	**ORCiD**[9]
First name	Y	Y	Possible	Possible	Possible	Possible
Surname	Y	Y	Possible	Possible	Possible	Possible
Email	Y	Y	Y	Y	Public email	Possible
commonName	Y	Y	N	N	N	N
uid	Y	Y	N	N	N	N
eduPersonOrcid	Possible	Possible	N	N	N	Y
eduPersonPrincipalName/ samlSubjectID	Y	Y	N Must be generated	N Must be generated	N Must be generated	N Must be generated
Home organisation	Y	Only VH	N	N	N	Possible, but not practical
IdP Name/EntityID	Y	Only VH	Google	Microsoft	GitHub	ORCiD
Affiliation/Scoped Affiliation	Y	Y	N	N	N	N
eduPersonAssurance REFEDS RAF[10]	Rolling out	Rolling out	N	N	N	N
authenticationMethod REFEDS MFA[11]	Rolling out	Rolling out	N	N	N	N

[7] The details (other than email) asserted by social identity providers (Google, Microsoft and GitHub) are self-asserted values.
[8] For GitHub, values might only be available if the user has made them public in their profile.
[9] Attributes marked as *Possible* can be released by ORCiD but are not release by default due to user privacy settings. Values may be user self-asserted. Encouraging users to configure their account to release these attributes is a violation of the ORCiD rules.
[10] REFEDS RAF Identity assurance profiles are a set of standards met by an identity assertion, based on the identity provider's identity management processes, the strength of its binding, etc. The values asserted indicate the strength of the processes used to identify the user at the time of user registration and credential issuance. See REFEDS for more details: https://refeds.org/wp-content/uploads/2023/12/RAF-2.0-Final-version.pdf
[11] Indicates the type of authentication credential in use and the strength of the user binding to the credential. See REFEDS MFA Profile https://refeds.org/profile/mfa

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